

# Mechanics

Lecture notes, worked examples and tutorial questions from past Singapore Junior Physics Olympiad papers

WORKED SOLUTIONS

Name \_\_\_\_\_

Class \_\_\_\_\_

Start date \_\_\_\_\_

Target finish \_\_\_\_\_

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# 1 Kinematics

Worked solutions to the examples in Chapter 1.

## 1.1 VECTORS AND MEASUREMENT

### EXAMPLE 1.1 SJPO 2026 Q1

#### Question 1

Which of the following is a unit vector? ( $\hat{i}, \hat{j}, \hat{k}$  are standard unit vectors in the  $x, y, z$  directions respectively.)

- (a)  $\hat{i} + \hat{j}$
- (b)  $\hat{i} \cdot \hat{i}$
- (c)  $\hat{i} \times \hat{i}$
- (d)  $(\hat{i} \times \hat{j}) \times \hat{k}$
- (e)  $(\hat{i} \times \hat{j}) \times \hat{i}$

- 1 (a)  $|\hat{i} + \hat{j}| = \sqrt{1^2 + 1^2} = \sqrt{2} \neq 1$ . Not a unit vector.
- 2 (b)  $\hat{i} \cdot \hat{i} = 1$  is a **scalar**, not a vector.
- 3 (c)  $\hat{i} \times \hat{i} = \vec{0}$ . (d)  $(\hat{i} \times \hat{j}) \times \hat{k} = \hat{k} \times \hat{k} = \vec{0}$ .
- 4 (e)  $(\hat{i} \times \hat{j}) \times \hat{i} = \hat{k} \times \hat{i} = \hat{j}$ , which has magnitude 1. **Answer (e)**

**Answer: (e)**

### EXAMPLE 1.2 SJPO 2018 Q50

50. Suppose that you own an unmarked gold bar. When you measured its mass, the scale that you used gives a reading of 0.998 kg. The uncertainty of the scale is given as  $\pm 0.005$  kg. However, the very next day, you found that there are 3 unmarked gold bars in the place where you had left your own gold bar. When you measured their mass **using the same scale**, you found that:

gold bar a = 0.991 kg  
gold bar b = 1.002 kg  
gold bar c = 1.005 kg

Which of these gold bars **could possibly** be your original unmarked gold bar?

- A. None of them
- B. Gold bar b only
- C. Either gold bar a or gold bar b only
- D. Either gold bar b or gold bar c only
- E. Any of them

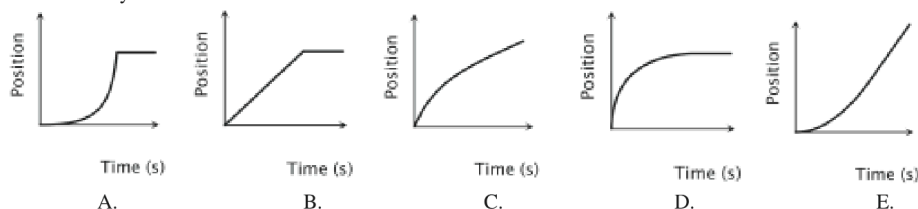
- 1 Original bar: true mass in  $[0.993, 1.003]$  kg.
- 2 Bar a:  $[0.986, 0.996]$ ; bar b:  $[0.997, 1.007]$ ; bar c:  $[1.000, 1.010]$  kg.
- 3 Every range overlaps  $[0.993, 1.003]$ , so any of them could be the original bar. **Answer E**

**Answer: E**

## 1.2 MOTION IN A STRAIGHT LINE

### EXAMPLE 1.3 SJPO 2018 Q10

10. The position of an object moving along a linear track is plotted as a function of time. It started from rest and underwent a positive acceleration for some time, followed by a constant velocity. Which of the following graphs correctly shows this situation?



- 1 Gradient of an  $s-t$  graph = velocity. Starting from rest means the gradient is zero at  $t = 0$ .
- 2 Positive acceleration: the gradient increases, so the curve bends upward. Then constant velocity: a straight line with constant positive gradient.
- 3 Only graph E is flat at the start, curves upward, then becomes straight. **Answer E**

**Answer: E**

### EXAMPLE 1.4 SJPO 2018 Q13

13. Travelling with an initial speed of 70 km/h, a car accelerates at 6000 km/h<sup>2</sup> along a straight road. How long will it take to reach a speed of 120 km/h?
- A. 30 s  
B. 45 s  
C. 60 s  
D. 70 s  
E. 180 s

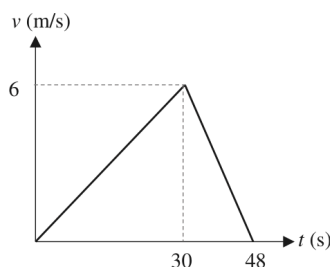
- 1 Keep units in km and h:  $v = u + at \Rightarrow 120 = 70 + 6000 t$
- 2  $t = \frac{50}{6000} \text{ h} = \frac{1}{120} \text{ h} = 30 \text{ s}$ . **Answer A**

**Answer: A**

### EXAMPLE 1.5 SJPO 2018 Q16–17

Q 16 and 17 refer to the following.

A car travels along a straight road with the speed shown by the  $v-t$  graph.



16. What is the acceleration of the car from  $t=30$  to  $t=48$  s?
- A.  $-54 \text{ m/s}^2$   
B.  $48 \text{ m/s}^2$   
C.  $-3.0 \text{ m/s}^2$   
D.  $3.0 \text{ m/s}^2$   
E.  $-0.33 \text{ m/s}^2$
17. What is the total displacement of the car after 48 s?
- A. 36 m  
B. 48 m  
C. 144 m  
D. 180 m  
E. 210 m

1 Q16:  $a = \frac{\Delta v}{\Delta t} = \frac{0 - 6}{48 - 30} = -0.33 \text{ m s}^{-2}$ . **Answer E**

2 Q17: displacement = area under the graph =  $\frac{1}{2}(48)(6) = 144 \text{ m}$ . **Answer C**

**Answer: Q16 E, Q17 C**

### EXAMPLE 1.6 SJPO 2024 Q3

3. A particle travels in a straight path with constant acceleration. The particle has velocity  $u$  at the start of its path and velocity  $v$  at the end of its path. What is the velocity of the particle at the midpoint of its path?

- A.  $\frac{1}{2}(v - u)$
- B.  $\frac{1}{2}(u + v)$
- C.  $\sqrt{uv}$
- D.  $\sqrt{u^2 + v^2}$
- E.  $\sqrt{\frac{1}{2}(u^2 + v^2)}$

1 Let the total length be  $2s$  and the midpoint velocity  $v_m$ . Apply  $v^2 = u^2 + 2as$  to each half:

2 
$$v_m^2 - u^2 = 2as \quad v^2 - v_m^2 = 2as$$

3 Equating:  $2v_m^2 = u^2 + v^2 \Rightarrow v_m = \sqrt{\frac{1}{2}(u^2 + v^2)}$ . **Answer E**

**Answer: E**

### EXAMPLE 1.7 SJPO 2022 Q1-3

For questions 1-3.

The road is level, and a driver is driving at the speed limit of 50 km/h (i.e. 14 m/s) when he sees the traffic light ahead turns yellow. He does not want to go over the speed limit and wants to enter the intersection before the light turns red. Assume the magnitude of the car's maximum deceleration rate is  $3.0 \text{ m/s}^2$  and the human reaction time is 1.0 s.

For questions 1 and 2: Can he avoid running the red light if the distance to the intersection and the duration of the yellow light are as follows:

1. 50 m and 3 s

- a) Yes, by applying his brake and comes to a stop before the intersection only.
- b) Yes, by continuing to drive at the speed limit only.
- c) Yes, both a) and b) work.
- d) No, both a) and b) do not work, and he will run the red light.

2. 40 m and 2 s

- a) Yes, by continuing to drive at the speed limit only.
- b) Yes, by applying his brake and comes to a stop before the intersection only.
- c) Yes, both a) and b) work.
- d) No, both a) and b) do not work, and he will run the red light.

3. What is the minimum duration for the yellow light so that the law-abiding driver can avoid running the red light, either by continuing to drive at the speed limit or by applying his brake and comes to a stop before the intersection, when he is approaching one?

- a) 6.33s
- b) 2.33s
- c) 4.33s
- d) 3.33s
- e) 5.33s

1 Stopping distance:  $s_{\text{stop}} = (14)(1) + \frac{14^2}{2(3)} = 14 + 32.7 = 46.7 \text{ m}$ .

- 2 Driving on at  $14 \text{ m s}^{-1}$  for the yellow time  $T$  covers  $14T$ .
- 3 Q1 (50 m, 3 s):  $14 \times 3 = 42 < 50$ , so he cannot clear it. But  $46.7 < 50$ , so he can stop. **Answer (a)**
- 4 Q2 (40 m, 2 s):  $14 \times 2 = 28 < 40$  and  $46.7 > 40$ . Neither works. **Answer (d)**
- 5 Q3: there must be no distance  $d$  at which both options fail. Braking works for  $d \geq 46.7 \text{ m}$  and driving on works for  $d \leq 14T$ . For no gap we need  $14T \geq 46.7$ , so  $T \geq 3.33 \text{ s}$ . **Answer (d)**

**Answer: Q1 a, Q2 d, Q3 d**

**About the answer key.** The published answer key gives 5.67 s, the total time taken to stop. That is not what Q3 asks. The minimum yellow time is set by the gap between the two strategies, which gives 3.33 s.

### 1.3 RELATIVE MOTION

#### EXAMPLE 1.8 SJPO 2024 Q2

2. Chris is seated on a northbound train when he sees a southbound train approaching him with speed  $u$ . At the same time, he sees a tree moving away from him at speed  $v$ . At what speed does a passenger seated on the southbound train observe this tree to be moving?

- A.  $u + v$
- B.  $u - v$
- C.  $u - 2v$
- D.  $2u + v$
- E.  $2u - v$

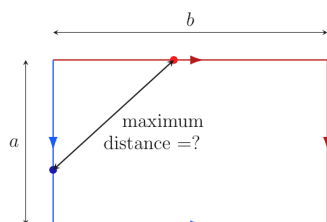
- 1 The tree is stationary, so the northbound train's ground speed is  $v$  (north).
- 2 Relative speed of the trains is  $u$ :  $v_S + v = u \Rightarrow v_S = u - v$  (south).
- 3 A passenger on the southbound train sees the tree move at  $u - v$ . **Answer B**

**Answer: B**

#### EXAMPLE 1.9 SJPO 2026 Q3

##### Question 3

Two runners start on the same corner of a rectangular track (e.g. upper-left corner) of sides  $a$  and  $b$ , with  $a < b$ . They begin running towards the opposite corner of the track at the same speed through two different paths around the track. What is the maximum distance reached between the two runners?



- (a)  $a$
  - (b)  $b$
  - (c)  $a\sqrt{2}$
  - (d)  $b\sqrt{2}$
  - (e)  $\sqrt{a^2 + b^2}$
- 1 Phase 1: runner  $a$  goes down the short side, runner  $b$  goes along the long side. They move perpendicular to each other, so their separation grows.
  - 2 Phase 2: after time  $a/v$  both runners move horizontally at the same speed, so their separation stays constant.

- 3 Phase 3: runner  $b$  turns downward and they close in. Maximum separation is at the start of phase 2: horizontal gap  $a$  and vertical gap  $a$ .
- 4  $d_{\max} = \sqrt{a^2 + a^2} = a\sqrt{2}$ . **Answer (c)**

**Answer: (c)**

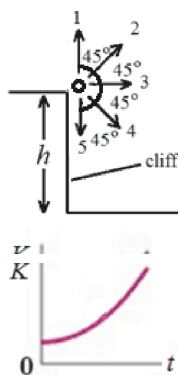
## 1.4 PROJECTILE MOTION

### EXAMPLE 1.10 SJPO 2022 Q6–7

For questions 6 and 7.

A ball is launched with a non-zero velocity  $v_0$  in one of the five directions from the edge of a cliff. Direction 1 is vertically up and direction 5 is vertically down, and direction 3 is rightward horizontally while directions 2 (4) are mid-way between 1 and 3 (3 and 5).

The block's kinetic energy  $K$  varies with time as shown. Note that the gradient of the  $K$ - $t$  graph is **zero** at  $t = 0$ .



6. Which is the direction of the initial velocity of the ball?

- a) 4
- b) 2
- c) 3
- d) 1
- e) 5

7. Ignoring air resistance, the time taken for the ball to land at the bottom of the cliff is \_\_\_\_.

- a)  $\frac{v_0}{g} \left( 1 + \sqrt{1 + \frac{2gh}{v_0^2}} \right)$
- b)  $\frac{v_0}{\sqrt{2}g} \left( 1 + \sqrt{1 + \frac{4gh}{v_0^2}} \right)$
- c)  $\sqrt{2gh}$
- d)  $\frac{v_0}{\sqrt{2}g} \left( \sqrt{1 + \frac{4gh}{v_0^2}} - 1 \right)$
- e)  $\frac{v_0}{g} \left( \sqrt{1 + \frac{2gh}{v_0^2}} - 1 \right)$

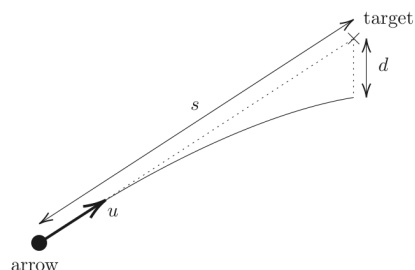
- 1  $K = \frac{1}{2}m(v_x^2 + v_y^2)$  with  $v_x$  constant and  $\frac{dv_y}{dt} = -g$ . So  $\frac{dK}{dt} = mv_y \frac{dv_y}{dt} = -mg v_y$ .
- 2  $\frac{dK}{dt} = 0$  at  $t = 0$  means  $v_y(0) = 0$ : the launch is horizontal, direction 3. **Q6 answer (c)**
- 3 Q7: horizontal launch, so vertically it is pure free fall:  $h = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{2h/g}$ .
- 4 Option (c) is printed as  $\sqrt{2gh}$ , which has units of speed, not time. The intended answer is (c),  $\sqrt{2h/g}$ , which is independent of  $v_0$ . **Q7:  $t = \sqrt{2h/g}$**

**Answer: Q6 c; Q7  $\sqrt{2h/g}$**

**About the answer key.** Q7 has a misprint: none of the printed options is dimensionally correct. The answer key also identifies (c) as the intended option.

### EXAMPLE 1.11 SJPO 2024 Q11

11. Paul fires an arrow at initial speed  $u$  towards a target a distance  $s$  away from him. He unfortunately makes the mistake of aiming directly at the target, and as a result, his arrow misses the target by a vertical distance  $d$ . What is  $d$ ?



- A.  $\frac{u^2}{2g}$
- B.  $\frac{gs^2}{2u^2}$
- C.  $\frac{2gs^2}{u^2}$
- D.  $\left(\frac{u^2}{2g} - 1\right)s$
- E.  $\left(\frac{gs}{2u^2} - 1\right)s$

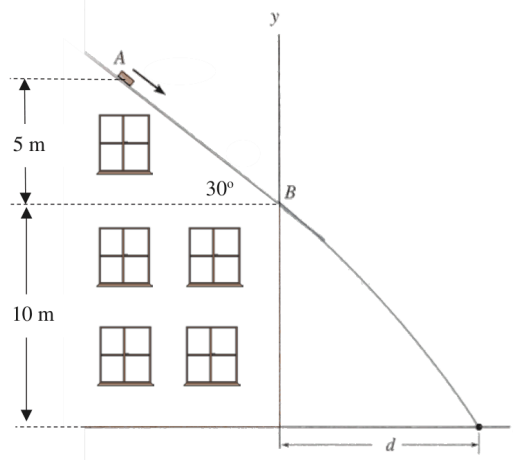
- 1 In the free-fall frame the arrow moves in a straight line at speed  $u$  along the line of sight, reaching the target's line after  $t = s/u$ .
- 2 In the ground frame everything has also fallen  $\frac{1}{2}gt^2$ :  $d = \frac{1}{2}g\left(\frac{s}{u}\right)^2 = \frac{gs^2}{2u^2}$ . **Answer B**

**Answer: B**

### EXAMPLE 1.12 SJPO 2018 Q18

18. A 2 kg block slides down a smooth roof which is angled at  $30^\circ$  to the horizontal. When it reaches A, it has a speed of 2.0 m/s. It leaves the surface of the roof at B and strikes the ground at a distance  $d$  from the wall of the building. What is the distance  $d$ ?

- A. 3.96 m
- B. 8.66 m
- C. 8.78 m
- D. 17.3 m
- E. 17.8 m

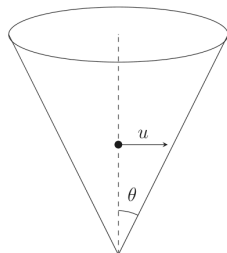


- 1 Energy from A to B:  $\frac{1}{2}mu^2 = \frac{1}{2}m(2.0)^2 + mg(5) \Rightarrow u = \sqrt{4 + 98} = 10.1 \text{ m s}^{-1}$
- 2 At B the velocity is  $10.1 \text{ m s}^{-1}$  at  $30^\circ$  below horizontal:  $u_x = 8.75$ ,  $u_y = 5.05 \text{ m s}^{-1}$  (down).
- 3 Vertically (down +):  $10 = 5.05t + 4.9t^2 \Rightarrow t = 1.00 \text{ s}$
- 4  $d = u_x t = 8.75 \times 1.00 = 8.78 \text{ m}$ . **Answer C**

**Answer: C**

**EXAMPLE 1.13** SJPO 2026 Q13**Question 13**

A particle is held above the tip of an inverted hollow cone of half-angle  $\theta$ . The particle is projected with a horizontal speed  $u$  such that it bounces back and forth indefinitely between the two sides of the cone symmetrically. What is the period of its motion? Neglect air resistance and assume all collisions are elastic.



- (a)  $\frac{u}{g} \sin \theta$
- (b)  $\frac{2u}{g} \sin \theta$
- (c)  $\frac{4u}{g} \sin \theta$
- (d)  $\frac{u}{g} \tan \theta$
- (e)  $\frac{4u}{g} \tan \theta$

- 1 To bounce back and forth indefinitely, the particle must retrace its path. So it must hit each wall **perpendicularly**.
- 2 The wall makes angle  $\theta$  with the vertical, so its normal makes angle  $\theta$  below the horizontal. We need  $\frac{v_y}{v_x} = \tan \theta \Rightarrow gt = u \tan \theta$ .
- 3  $t = \frac{u}{g} \tan \theta$ . One period is: out to wall A, back to the start, out to wall B, back again, which is 4 such flights.
- 4  $T = \frac{4u}{g} \tan \theta$ . **Answer (e)**

**Answer: (e)**

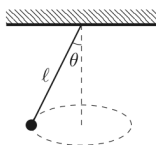
# 2 Newton's Laws and Dynamics

Worked solutions to the examples in Chapter 2.

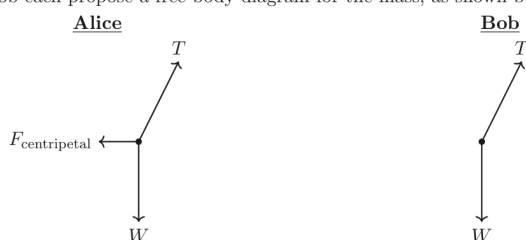
## 2.1 NEWTON'S LAWS AND FREE-BODY DIAGRAMS

### EXAMPLE 2.1 SJPO 2024 Q7

7. In a “circular pendulum”, a mass  $m$  is suspended by a string of length  $\ell$  and travels in a circular path at constant speed.



Alice and Bob each propose a free-body diagram for the mass, as shown below.



Whose free-body diagram is correct, and why?

- Alice, as the mass must be in equilibrium.
- Alice, as the centripetal force is a fictitious force that must be included.
- Bob, as tension and weight are the only forces on the mass.
- Bob, as tension and weight form an action-reaction pair.
- Both are valid diagrams depending on the frame of reference chosen.

- In the ground frame the only forces on the bob are the tension  $T$  and the weight  $W$ .
- Their resultant points horizontally towards the centre. This resultant is the centripetal force, so drawing it again double counts. The bob accelerates, so it is *not* in equilibrium.
- $T$  and  $W$  are not a third-law pair: they are different types of force acting on the same body.

**Answer C**

**Answer: C**

### EXAMPLE 2.2 SJPO 2024 Q4

4.  $n$  identical blocks (where  $n > 2$ ) are placed side-by-side in a horizontal row on frictionless ground. The combined row of blocks is pushed from the left with a force  $F$ . What is the force between the two rightmost blocks?

- $F$
- $nF$
- $\frac{F}{n}$
- $\frac{F}{n-1}$
- $\frac{(n-1)F}{n}$

- System:  $a = \frac{F}{nm}$ .
- Isolate the rightmost block. The only horizontal force on it is the push from its neighbour:  $N = ma = \frac{F}{n}$ . **Answer C**

**Answer: C**

**EXAMPLE 2.3** SJPO 2018 Q2

2. A tiny spherical raindrop of diameter  $D = 0.1$  mm experiences a linear drag force while falling at speed  $v$ , given by  $F_{\text{drag}} = cv$ , where  $c = 1.55 \times 10^{-6}$  N s m<sup>-1</sup>. What is its terminal speed?
- A. 1.7 mm / s  
B. 3.3 mm / s  
C. 8.5 mm / s  
D. 26 mm / s  
E. 33 mm / s

- 1 At terminal speed:  $cv_T = mg$ , with  $m = \rho \cdot \frac{4}{3}\pi r^3 = 1000 \cdot \frac{4}{3}\pi(5 \times 10^{-5})^3 = 5.24 \times 10^{-10}$  kg.
- 2  $v_T = \frac{mg}{c} = \frac{(5.24 \times 10^{-10})(9.80)}{1.55 \times 10^{-6}} = 3.3 \times 10^{-3} \text{ m s}^{-1} = 3.3 \text{ mm s}^{-1}$ . **Answer B**

**Answer: B**

**2.2 FRICTION**

**EXAMPLE 2.4** SJPO 2026 Q6

**Question 6**

A block is released from rest on an inclined ramp, and slides down the ramp with acceleration  $a > 0$ . The ramp has coefficients of static and kinetic friction  $\mu_s$  and  $\mu_k$  respectively. What is the smallest possible value of  $a$ ?

- (a)  $(1 - \mu_k)g$   
(b)  $(\mu_s - \mu_k)g$   
(c)  $\frac{\mu_s}{\sqrt{1 + \mu_s^2}}g$   
(d)  $\frac{\mu_k}{\sqrt{1 + \mu_s^2}}g$   
(e)  $\frac{\mu_s - \mu_k}{\sqrt{1 + \mu_s^2}}g$

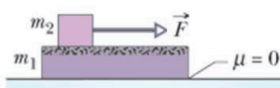
- 1  $N = mg \cos \theta$ ; once sliding,  $a = g(\sin \theta - \mu_k \cos \theta)$ , which increases with  $\theta$ .
- 2 It just starts to slide when  $mg \sin \theta = \mu_s mg \cos \theta$ , i.e.  $\tan \theta = \mu_s$ .
- 3 Then  $\sin \theta = \frac{\mu_s}{\sqrt{1 + \mu_s^2}}$ ,  $\cos \theta = \frac{1}{\sqrt{1 + \mu_s^2}}$ , so  $a_{\min} = \frac{\mu_s - \mu_k}{\sqrt{1 + \mu_s^2}}g$ . **Answer (e)**

**Answer: (e)**

**EXAMPLE 2.5** SJPO 2022 Q12–13

For questions 12 and 13

12. A slab of mass  $m_1$  rests on a frictionless floor, and a block of mass  $m_2$  rests on top of the slab. Between the block and slab, the coefficient of static friction is  $\mu_s$  and the coefficient of kinetic friction is  $\mu_k$ . A horizontal force of magnitude  $F$  begins to pull directly on the block, as shown. The condition for the two masses to move as one piece is, \_\_\_\_, where  $M = m_1 + m_2$ .



- a)  $\mu_s \leq \frac{m_1}{m_2} \frac{F}{Mg}$
- b)  $\mu_s \leq \frac{m_2}{m_1} \frac{F}{Mg}$
- c)  $\mu_s \geq \frac{m_1}{m_2} \frac{F}{Mg}$
- d)  $\mu_s \geq \frac{m_2}{m_1} \frac{F}{Mg}$

e) The two masses will always move as one piece since there is friction between them and no friction between the slab and the floor.

Hint: The kinetic friction on an object is given by  $\mu_k N$  where  $N$  is the normal force on the object. The maximum static friction on an object is given by  $\mu_s N$ .

13. Suppose  $m_1 = 50$  kg and  $m_2 = 20$  kg,  $\mu_s = 0.55$ ,  $\mu_k = 0.40$  and  $F = 140$  N, what is the resulting acceleration (in  $\text{m} \cdot \text{s}^{-2}$ ) of the block? Take  $g = 10$  N/kg.

- a) 7.0
- b) 3.0
- c) 3.3
- d) 1.6
- e) 2.0

- 1 Assume they move together:  $a = \frac{F}{M}$  with  $M = m_1 + m_2$ .
- 2 The slab is driven only by friction from the block:  $f = m_1 a = \frac{m_1 F}{M}$ .
- 3 No slipping requires  $f \leq \mu_s m_2 g$ , so  $\mu_s \geq \frac{m_1}{m_2} \frac{F}{Mg}$ . **Q12 answer (c)**
- 4 Q13:  $\frac{m_1}{m_2} \frac{F}{Mg} = \frac{50}{20} \cdot \frac{140}{70 \times 10} = 0.50 \leq 0.55$ , so they stick. Then  $a = \frac{140}{70} = 2.0 \text{ m s}^{-2}$ . **Q13 answer (e)**

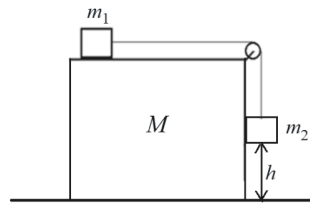
**Answer: Q12 c, Q13 e**

## 2.3 CONNECTED BODIES AND PULLEYS

### EXAMPLE 2.6 SJPO 2022 Q17–18

For questions 17–21

A block of mass  $m_1$  is placed on top of a block of mass  $M$ . Holding block  $m_1$  stationary, it is joined by a string which passes over the pulley to a block of mass  $m_2$  hanging by the side of (and in contact with) block  $M$ , and is at height  $h$  above the floor. We will assume that the string is inextensible and massless, and all the contacts between objects in relative motion are frictionless. Furthermore, we will be concerned with the part of the motion before  $m_1$  reaches the pulley.



For questions 17 and 18, assume that  $M$  is fixed to the floor, and the pulley is massless. After  $m_1$  is released from rest, both  $m_1$  and  $m_2$  move smoothly with a constant acceleration and the string remains taut.

17. What is the acceleration of blocks  $m_1$  and  $m_2$ ?

- a)  $\frac{m_1}{m_1+m_2}g$
- b)  $\frac{m_2}{m_1+m_2}g$
- c)  $g$
- d)  $\frac{m_1+m_2}{m_1}g$
- e)  $\frac{m_1+m_2}{m_2}g$

18. What is the time taken for  $m_2$  to reach the floor after  $m_1$  is released from rest?

- a)  $\sqrt{\frac{2h(m_1+m_2)}{m_1g}}$
- b)  $\sqrt{\frac{2h(m_1+m_2)}{m_2g}}$
- c)  $\sqrt{\frac{2h}{g}}$
- d)  $\sqrt{\frac{2h m_1}{(m_1+m_2)g}}$
- e)  $\sqrt{\frac{2h m_2}{(m_1+m_2)g}}$

① Driving force along the string is  $m_2g$ , and the total mass moved is  $m_1 + m_2$ :  $a = \frac{m_2}{m_1 + m_2}g$ . **Q17 answer (b)**

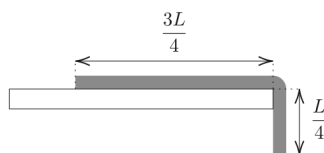
② From rest,  $h = \frac{1}{2}at^2 \Rightarrow t = \sqrt{\frac{2h}{a}} = \sqrt{\frac{2h(m_1 + m_2)}{m_2g}}$ . **Q18 answer (b)**

**Answer: Q17 b, Q18 b**

**EXAMPLE 2.7** SJPO 2024 Q5–6

The following information applies to both questions 5 and 6.

A uniform chain of length  $L$  is placed on a table, with a length  $\frac{L}{4}$  hanging over the edge of the table. When the chain is released from rest, it begins sliding off the table.



5. Suppose that the tabletop is frictionless. What is the initial acceleration of the chain?

- A.  $\frac{g}{4}$
- B.  $\frac{g}{3}$
- C.  $\frac{2g}{3}$
- D.  $\frac{3g}{4}$
- E.  $g$

6. Suppose now that the tabletop has friction. Given that the chain still slides off the table upon release, what is the maximum possible coefficient of static friction of the tabletop?

- A.  $\frac{1}{4}$
- B.  $\frac{1}{3}$
- C.  $\frac{2}{3}$
- D.  $\frac{3}{4}$
- E. 1

① Q5: driving force = weight of the hanging quarter =  $\frac{1}{4}mg$ , and the mass accelerated is  $m$ :  $a = \frac{g}{4}$ .

**Answer A**

② Q6: the part on the table has normal force  $\frac{3}{4}mg$ . To slide we need  $\frac{1}{4}mg > \mu_s \cdot \frac{3}{4}mg$ .

③  $\mu_s < \frac{1}{3}$ , so the maximum is  $\mu_s = \frac{1}{3}$ . **Answer B**

**Answer: Q5 A, Q6 B**

## 2.4 ACCELERATING FRAMES AND FICTITIOUS FORCES

### EXAMPLE 2.8 SJPO 2018 Q3

3. A plumb line is held steady while being carried in a moving train. The mass of the plumb bob is  $m$ , and the train is accelerating in the forward direction at  $0.5g$ . What is the tension in the string?

- A.  $0.5\,mg$
- B.  $1.12\,mg$
- C.  $1.22\,mg$
- D.  $2\,mg$
- E. Cannot be determined from the given information.

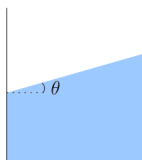
① Vertically:  $T \cos \theta = mg$ . Horizontally:  $T \sin \theta = m(0.5g)$ .

② Square and add:  $T = mg\sqrt{1^2 + 0.5^2} = 1.12\,mg$ . (In the train frame:  $T = mg_{\text{eff}}$ ) **Answer B**

**Answer: B**

**EXAMPLE 2.9** SJPO 2024 Q8

8. On board a shaky bus, Galen observes a glass of water and notices its surface to be tilted towards the right by an angle  $\theta = 35^\circ$ . What is the acceleration of the bus, and in which direction?



- A.  $5.6 \text{ m s}^{-2}$  towards the left
- B.  $5.6 \text{ m s}^{-2}$  towards the right
- C.  $6.9 \text{ m s}^{-2}$  towards the left
- D.  $6.9 \text{ m s}^{-2}$  towards the right
- E.  $9.8 \text{ m s}^{-2}$  downwards

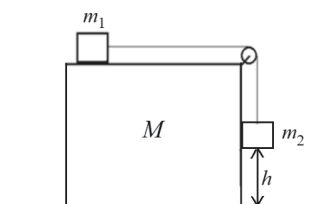
- 1 The surface is perpendicular to  $\vec{g}_{\text{eff}} = \vec{g} - \vec{a}_0$ . The water piles up on the right, so  $\vec{g}_{\text{eff}}$  tilts to the right and  $\vec{a}_0$  points **left**.
- 2  $\tan \theta = \frac{a_0}{g} \Rightarrow a_0 = 9.8 \tan 35^\circ = 6.9 \text{ m s}^{-2}$  to the left. **Answer C**

**Answer: C**

**EXAMPLE 2.10** SJPO 2022 Q20

For questions 17- 21

A block of mass  $m_1$  is placed on top of a block of mass  $M$ . Holding block  $m_1$  stationary, it is joined by a string which passes over the pulley to a block of mass  $m_2$  hanging by the side of (and in contact with) block  $M$ , and is at height  $h$  above the floor. We will assume that the string is inextensible and massless, and all the contacts between objects in relative motion are frictionless. Furthermore, we will be concerned with the part of the motion before  $m_1$  reaches the pulley.



For questions 19 and 20, all assumptions remain except that  $M$  is no longer fixed to the floor but can move without friction on the floor.

20. By applying a constant rightward force on  $M$ , both  $m_1$  and  $m_2$  can remain at rest relative to  $M$  without having to hold them. This force is \_\_\_\_.

- a)  $\frac{m_1}{(M+m_1+m_2)m_2}g$
- b)  $\frac{m_2}{(M+m_1+m_2)m_1}g$
- c)  $\frac{(M+m_1+m_2)m_1}{m_2}g$
- d)  $\frac{(M+m_1+m_2)m_2}{m_1}g$
- e) none of the above

- 1 Everything moves together, so  $a = \frac{F}{M + m_1 + m_2}$  to the right.
- 2 In  $M$ 's frame, a fictitious force  $m_1 a$  acts on  $m_1$  to the **left**, away from the pulley. For  $m_1$  to stay at rest:  $T = m_1 a$ . For  $m_2$  vertically:  $T = m_2 g$ .
- 3  $m_1 a = m_2 g \Rightarrow a = \frac{m_2}{m_1} g$ , so  $F = \frac{(M + m_1 + m_2) m_2}{m_1} g$ . **Answer (d)**

**Answer: d**

# 3 Forces in Equilibrium

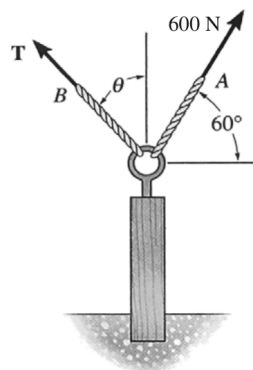
Worked solutions to the examples in Chapter 3.

## 3.1 EQUILIBRIUM OF CONCURRENT FORCES

### EXAMPLE 3.1 SJPO 2018 Q12

12. A wooden peg is to be pulled out of the ground using two ropes A and B. Rope A is subject to a force of 600 N at  $60^\circ$  to the horizontal. Rope B is pulled at a fixed angle  $\theta$  to the vertical. If the resultant force acting on the post is to be 1600 N vertically, what should be the force  $T$  on rope B?

- A. 1121 N
- B. 1334 N
- C. 1400 N
- D. 1924 N
- E. 2040 N



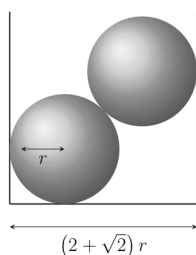
- 1 Horizontal components cancel:  $T \sin \theta = 600 \cos 60^\circ = 300$  N.
- 2 Vertical components sum to 1600 N:  $T \cos \theta = 1600 - 600 \sin 60^\circ = 1080$  N.
- 3  $T = \sqrt{300^2 + 1080^2} = 1121$  N. **Answer A**

**Answer: A**

### EXAMPLE 3.2 SJPO 2026 Q11

#### Question 11

A box of base width  $(2 + \sqrt{2})r$  is placed on a horizontal ground. Two identical spheres of mass  $m$  and radius  $r$  are packed inside it. What is the force exerted between the two spheres? Assume all surfaces are smooth.



- (a)  $mg$
- (b)  $2mg$
- (c)  $\frac{mg}{2}$
- (d)  $mg\sqrt{2}$
- (e)  $\frac{\sqrt{2}}{2}mg$

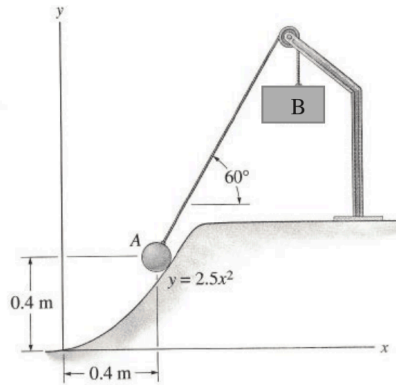
- 1 Centres are at  $x = r$  and  $x = (2 + \sqrt{2})r - r$ , so their horizontal separation is  $\sqrt{2}r$ . The distance between centres is  $2r$ .
- 2 Angle of the line of centres to the horizontal:  $\cos \theta = \frac{\sqrt{2}r}{2r} \Rightarrow \theta = 45^\circ$ .

- 3 Upper sphere: walls push horizontally only, so the vertical part of  $N$  balances  $mg$ :  $N \sin 45^\circ = mg \Rightarrow N = \sqrt{2} mg$ . **Answer (d)**

**Answer: (d)**

### EXAMPLE 3.3 SJPO 2018 Q6

6. A 4 kg sphere rests on the smooth parabolic surface which is given by the equation  $y = 2.5x^2$ . The sphere just touches the surface at  $(x, y) = (0.4 \text{ m}, 0.4 \text{ m})$ . Determine the mass  $m_B$  of block B needed to hold the sphere in equilibrium.
- A. 4.14 kg  
B. 3.75 kg  
C. 3.58 kg  
D. 2.93 kg  
E. 2.14 kg



- 1 Slope:  $\frac{dy}{dx} = 5x = 2.0$  at  $x = 0.4$ , so  $\tan \phi = 2$ . The normal force makes angle  $\phi$  with the vertical:  
 $N_x = N \sin \phi$  (left),  $N_y = N \cos \phi$ .
- 2 Horizontal:  $T \cos 60^\circ = N \sin \phi$ . Vertical:  $T \sin 60^\circ + N \cos \phi = m_A g$ .
- 3 Eliminate  $N$ :  $T \left( \sin 60^\circ + \frac{\cos 60^\circ}{\tan \phi} \right) = m_A g \Rightarrow T(0.866 + 0.25) = 4g \Rightarrow T = 3.58g$ .
- 4 The pulley transmits  $T = m_B g$ , so  $m_B = 3.58 \text{ kg}$ . **Answer C**

**Answer: C**

## 3.2 MOMENTS AND CENTRE OF MASS

### EXAMPLE 3.4 SJPO 2018 Q5

5. A thin wire is bent into the form of a three-sided shape as shown below. Each segment has equal length  $l$ . The height of the centre of mass from the bottom of the shape is \_\_\_\_.
- A.  $l/2$   
B.  $2l/3$   
C.  $l/3$   
D.  $l/4$   
E.  $2l/5$



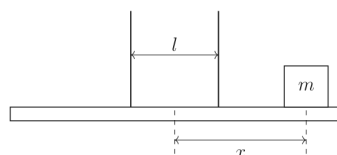
- 1 Each segment has mass  $m$ . The two vertical sides have CM at height  $l/2$ , and the bottom has its CM at height 0.
- 2  $y_{\text{cm}} = \frac{m(l/2) + m(l/2) + m(0)}{3m} = \frac{l}{3}$ . **Answer C**

**Answer: C**

### EXAMPLE 3.5 SJPO 2026 Q9

#### Question 9

A long uniform rod of mass  $M$  is held horizontal by two strings, attached symmetrically to the rod at distance  $l$  apart. A small block of mass  $m$  initially rests on the rod's centre. It is then slowly moved towards the right. At what distance  $x$  between the block and the rod's centre does one of the strings start to become slack?



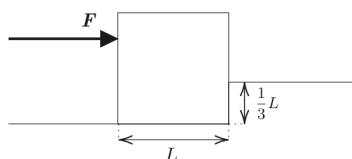
- (a)  $\frac{l}{2} \frac{M+m}{m}$
- (b)  $\frac{l}{2} \frac{M-m}{m}$
- (c)  $\frac{l}{2} \frac{M+m}{M}$
- (d)  $\frac{l}{2} \frac{m}{M+m}$
- (e)  $\frac{l}{2} \frac{M}{M+m}$

- 1 The left string goes slack when its tension reaches zero. The system is then supported by the right string alone, so its CM must lie directly below that string, at  $x = l/2$ .
- 2 Taking the rod's centre as origin:  $\frac{M(0) + mx}{M+m} = \frac{l}{2}$
- 3  $x = \frac{l}{2} \cdot \frac{M+m}{m}$ . **Answer (a)**

**Answer: (a)**

### EXAMPLE 3.6 SJPO 2024 Q9

9. A tired delivery worker has a cube-shaped box of mass  $M$  and side length  $L$  on the ground, beside a step of height  $\frac{1}{3}L$ . In an attempt to bring the box up the step, he applies a horizontal force  $F$  on the box. What is the smallest value of  $F$  required for him to succeed?



- A.  $Mg$
- B.  $\frac{1}{2}Mg$
- C.  $\frac{2}{3}Mg$
- D.  $\frac{3}{4}Mg$
- E.  $\frac{3}{2}Mg$

- 1 As the box starts to rise, it pivots about the step's corner and loses contact with the ground. Take moments about that corner.
- 2 For the minimum force, apply  $F$  at the top of the box. Its arm about the corner is  $L - \frac{1}{3}L = \frac{2}{3}L$ . The weight's arm is the horizontal distance to the CM,  $\frac{1}{2}L$ .
- 3  $F \cdot \frac{2}{3}L = Mg \cdot \frac{1}{2}L \Rightarrow F = \frac{3}{4}Mg$ . **Answer D**

**Answer: D**

# 4 Work, Energy and Power

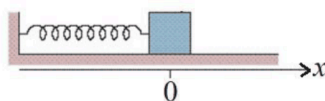
Worked solutions to the examples in Chapter 4.

## 4.1 WORK AND THE WORK-ENERGY THEOREM

### EXAMPLE 4.1 SJPO 2022 Q8

For questions 8 and 9.

A spring that obeys Hooke's law and a block of mass 2.0 kg are set up as shown. When the block is at  $x = 0$ , the spring is at its natural length. The block is pulled out to  $x = 5.0$  cm, and a force of magnitude 10 N is required to hold it there. Then the block is released. Ignore friction.



8. How much is the work done by the spring on the block to move it from  $x = 5.0$  cm to  $x = -3.0$  cm?

- a) 0.16J
- b) -0.16J
- c) 0.08J
- d) -0.08J
- e) cannot be determined as the spring constant is not given

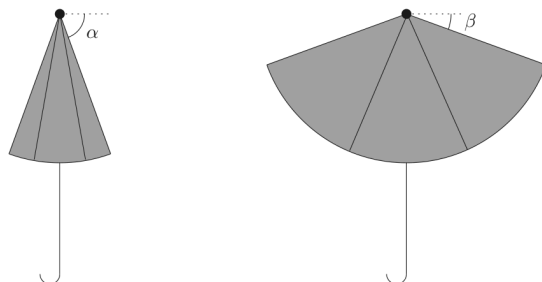
1  $k = \frac{F}{x} = \frac{10}{0.050} = 200 \text{ N m}^{-1}$ . The spring constant can be found, so (e) is wrong.

2  $W_{\text{spring}} = \frac{1}{2}k(x_1^2 - x_2^2) = \frac{1}{2}(200)(0.050^2 - 0.030^2) = +0.16 \text{ J}$ . **Answer (a)**

**Answer: a**

### EXAMPLE 4.2 SJPO 2024 Q13

13. The frame of an umbrella is constructed by attaching 6 uniform rods, each of mass  $m = 0.05$  kg and length  $l = 1.0$  m, to a common centre. The frame makes an angle  $\alpha = 80^\circ$  to the horizontal when the umbrella is closed, and an angle  $\beta = 20^\circ$  when the umbrella is open, as drawn below. (Note: Not all 6 rods are drawn due to perspective.)



How much work needs to be done to open the umbrella? Assume that the umbrella's central axis is kept vertical and stationary, and that the mass of its fabric is negligible compared to the mass of its frame.

- A. 0.95 J
- B. 1.1 J
- C. 1.9 J
- D. 2.3 J
- E. 2.9 J

- 1 Each rod hangs from the top at angle  $\alpha$  below horizontal, so its CM is  $\frac{l}{2} \sin \alpha$  below the hub.
- 2 Opening raises each CM by  $\frac{l}{2}(\sin 80^\circ - \sin 20^\circ) = 0.5(0.985 - 0.342) = 0.321 \text{ m}$ .

3  $W = 6mg \Delta h = 6(0.05)(9.8)(0.321) = 0.95 \text{ J}$ . **Answer A**

**Answer: A**

## 4.2 CONSERVATION OF ENERGY AND DISSIPATION

### EXAMPLE 4.3 SJPO 2018 Q19

19. Two adventurous Physics students, each weighing 60 kg, jump off a 43 m high bridge on a Bungee cord. The length of the cord is such that the students together will just touch the water and rebound. While Bungee cords become softer (more elastic) with increasing extension, for our calculations we can approximate the cord as having a constant stiffness of  $k = 330 \text{ N/m}$ , and we can ignore the height of the students and where the cord is tied on their body. What is the unstretched length of the cord?

- A. 10.0 m
- B. 17.5 m
- C. 25.5 m
- D. 32.5 m
- E. 40.0 m

1 At the water they are momentarily at rest, so all the GPE lost is stored in the cord:  $mgh = \frac{1}{2}kx^2$  with  $m = 120 \text{ kg}$ ,  $h = 43 \text{ m}$ .

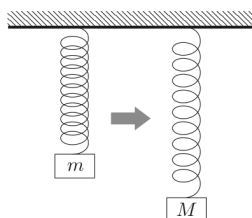
2  $x = \sqrt{\frac{2(120)(9.80)(43)}{330}} = 17.5 \text{ m}$ .

3 Unstretched length =  $43 - 17.5 = 25.5 \text{ m}$ . **Answer C**

**Answer: C**

### EXAMPLE 4.4 SJPO 2024 Q15

15. A mass  $m$  is suspended at rest by an ideal spring of force constant  $k$ . At this position, an extra mass is gently added so that the total mass is now  $M$ , before it is released from rest. The spring-mass system subsequently oscillates, but eventually settles to a rest due to damping by resistive forces. How much work is done in total by these resistive forces?



- A.  $-\frac{M^2g^2}{2k}$
- B.  $-\frac{M(M-m)g^2}{2k}$
- C.  $-\frac{(M^2-m^2)g^2}{2k}$
- D.  $-\frac{(M^2+m^2)g^2}{2k}$
- E.  $-\frac{(M-m)^2g^2}{2k}$

1 Initial and final rest positions:  $kx_0 = mg$  and  $kx_f = Mg$ . The mass is at rest at both, so  $W_{\text{res}} = \Delta U_{\text{total}}$ .

2  $\Delta U = \frac{1}{2}kx_f^2 - \frac{1}{2}kx_0^2 - Mg(x_f - x_0)$

3 Substituting  $x = \frac{\text{weight}}{k}$ :  $\Delta U = \frac{g^2}{2k} [(M^2 - m^2) - 2M(M - m)] = -\frac{(M - m)^2g^2}{2k}$ . **Answer E**

**Answer: E**

# 5 Momentum and Collisions

Worked solutions to the examples in Chapter 5.

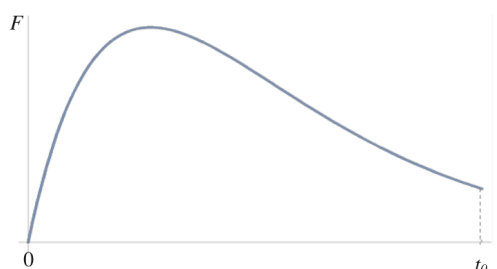
## 5.1 IMPULSE AND MOMENTUM

### EXAMPLE 5.1 SJPO 2018 Q15

15. The twitch in an arm muscle develops a force  $F$  which can be measured as a function of time  $t$ , given by  $F = F_0 \left( \frac{t}{T} \right) e^{-t/T}$ , where  $F_0$  and  $T$  are constants. The function is plotted in the figure. If the ratio  $t_0/T = 1$ , what is the impulse developed between time  $t = 0$  and time  $t = t_0$ ? (Hint: Make use of this integral.

$$\int_0^a x e^{-x} dx = 1 - (1+a)e^{-a}.)$$

- A.  $F_0(T(1/e) + t_0(1/e))$
- B.  $F_0(T(1/e) - t_0(1/e))$
- C.  $F_0(T(1 - 1/e) + t_0(1/e))$
- D.  $F_0(T(1 - 1/e) - t_0(1/e))$
- E. Indeterminate



- 1  $J = \int_0^{t_0} F_0 \frac{t}{T} e^{-t/T} dt$ . Let  $x = t/T$ , so  $dt = T dx$  and the limits become  $0 \rightarrow t_0/T = 1$ .
- 2  $J = F_0 T \int_0^1 x e^{-x} dx = F_0 T [1 - (1+1)e^{-1}] = F_0 T (1 - \frac{2}{e})$
- 3 Writing  $T(1 - \frac{2}{e}) = T(1 - \frac{1}{e}) - t_0(\frac{1}{e})$  (since  $t_0 = T$ ) gives option D. **Answer D**

**Answer: D**

**About the answer key.** The official solution derives exactly the expression in option D but then prints "ANSWER: E". This looks like a typo: the impulse is determinate, and it equals option D.

### EXAMPLE 5.2 SJPO 2022 Q4-5

For questions 4 and 5.

Two blocks, A and B, of masses  $m_A = 4.0$  kg and  $m_B = 20.0$  kg are in contact and are moving with velocity  $1.0 \text{ ms}^{-1}$  on a horizontal floor. The 36-N constant force is then applied to A for 5 seconds as shown. Ignore friction.



4. After 5 seconds, the momentum (in  $\text{kg} \cdot \text{m} \cdot \text{s}^{-1}$ ) of block B is \_\_\_\_.

- a) 170
- b) 204
- c) 120
- d) 150
- e) 240

5. After 5 seconds, the net work done (in J) on the block A is \_\_\_\_.

- a) 712.5
- b) 142.5
- c) 663
- d) 112.5
- e) 855

- 1 Impulse on the system:  $J = F\Delta t = 36 \times 5 = 180 \text{ N s}$ . Initial momentum  $= 24 \times 1.0 = 24 \text{ kg m s}^{-1}$ .
- 2 Final:  $24v = 24 + 180 \Rightarrow v = 8.5 \text{ m s}^{-1}$ . So  $p_B = 20 \times 8.5 = 170 \text{ kg m s}^{-1}$ . **Q4 answer (a)**
- 3 Q5:  $W_{\text{net,A}} = \Delta K_A = \frac{1}{2}(4.0)(8.5^2 - 1.0^2) = 142.5 \text{ J}$ . **Q5 answer (b)**

**Answer: Q4 a, Q5 b**

## 5.2 CONSERVATION OF MOMENTUM AND COLLISIONS

### EXAMPLE 5.3 SJPO 2018 Q14

14. Two blocks move towards each other on a smooth table with the velocities as shown. Block A has mass 5 kg and moves at 2 m/s to the right. Block B has mass 2 kg and moves at 5 m/s to the left. The coefficient of restitution  $e$  is the ratio of the final relative speed of separation to the initial relative speed of approach of two colliding objects. Take rightwards as positive. If  $e = 0.5$ , what is the velocity of block B after impact?
- A. -2.5 m/s  
B. -1.0 m/s  
C. 0 m/s  
D. 1.0 m/s  
E. 2.5 m/s



- 1 Momentum (right +):  $5(2) + 2(-5) = 0 = 5v_A + 2v_B$ .
- 2 Restitution: approach speed  $= 2 - (-5) = 7$ , so  $v_B - v_A = 0.5 \times 7 = 3.5$ .
- 3 Solving:  $v_A = -1.0 \text{ m s}^{-1}$ ,  $v_B = +2.5 \text{ m s}^{-1}$ . **Answer E**

**Answer: E**

### EXAMPLE 5.4 SJPO 2024 Q12

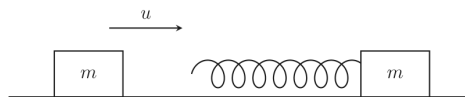
12. Bobo and Nana are initially at rest on icy frictionless ground. Bobo has mass  $M_1$  and Nana has mass  $M_2$ . Bobo grabs a snowball of mass  $m$  and throws it at Nana, who catches it. Bobo and Nana end up moving apart at speeds  $v_1$  and  $v_2$  respectively. What is the ratio  $\frac{v_2}{v_1}$ ?
- A.  $\frac{M_1}{M_2}$   
B.  $\frac{M_2}{M_1}$   
C.  $\frac{M_1}{M_2 + m}$   
D.  $\frac{M_2}{M_1 + m}$   
E.  $\frac{M_1 + m}{M_2 + m}$

- 1 Throw (Bobo + ball start at rest):  $mv = M_1v_1$ .
- 2 Catch:  $mv = (M_2 + m)v_2$ .
- 3  $\frac{v_2}{v_1} = \frac{M_1}{M_2 + m}$ . **Answer C**

**Answer: C**

**EXAMPLE 5.5** **SJPO 2026 Q7****Question 7**

A block of mass  $m$  is launched with speed  $u$  towards an ideal spring of force constant  $k$ , which is attached to another block of equal mass  $m$ . Both blocks are free to slide on a frictionless horizontal surface. What is the maximum compression of the spring?



- (a)  $\frac{u}{2} \sqrt{\frac{m}{k}}$
- (b)  $\frac{u\sqrt{2}}{2} \sqrt{\frac{m}{k}}$
- (c)  $\frac{u\sqrt{3}}{2} \sqrt{\frac{m}{k}}$
- (d)  $\frac{u\sqrt{6}}{2} \sqrt{\frac{m}{k}}$
- (e)  $u \sqrt{\frac{m}{k}}$

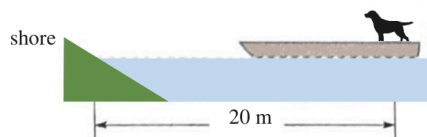
- 1 Maximum compression occurs at common velocity:  $mu = 2mv \Rightarrow v = u/2$ .
- 2 Energy:  $\frac{1}{2}mu^2 = \frac{1}{2}(2m)\left(\frac{u}{2}\right)^2 + \frac{1}{2}kx^2 \Rightarrow \frac{1}{2}kx^2 = \frac{1}{4}mu^2$
- 3  $x = u\sqrt{\frac{m}{2k}} = \frac{u\sqrt{2}}{2} \sqrt{\frac{m}{k}}$ . **Answer (b)**

**Answer: (b)**

**5.3 CENTRE OF MASS MOTION****EXAMPLE 5.6** **SJPO 2018 Q20**

20. A dog weighing 10 kg is standing on a flatboat so that he is 20 m from the shore. The boat weighs 40 kg and has uniform mass distribution. The dog walks 8.0 m on the boat towards the shore and stops. For this calculation, one can assume no friction or drag between the boat and the water. How far is the dog from the shore now?

- A. 12.0 m
- B. 13.6 m
- C. 14.0 m
- D. 16.8 m
- E. 29.8 m



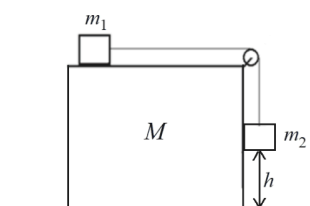
- 1 Take towards shore as positive. Dog and boat ground displacements  $\Delta x_d$ ,  $\Delta x_b$ , with  $\Delta x_d - \Delta x_b = 8.0$  m.
- 2 No horizontal external force, starting from rest:  $10 \Delta x_d + 40 \Delta x_b = 0 \Rightarrow \Delta x_b = -\frac{1}{4} \Delta x_d$ .
- 3  $\frac{5}{4} \Delta x_d = 8.0 \Rightarrow \Delta x_d = 6.4$  m. Distance from shore =  $20 - 6.4 = 13.6$  m. **Answer B**

**Answer: B**

**EXAMPLE 5.7** SJPO 2022 Q19

For questions 17- 21

A block of mass  $m_1$  is placed on top of a block of mass  $M$ . Holding block  $m_1$  stationary, it is joined by a string which passes over the pulley to a block of mass  $m_2$  hanging by the side of (and in contact with) block  $M$ , and is at height  $h$  above the floor. We will assume that the string is inextensible and massless, and all the contacts between objects in relative motion are frictionless. Furthermore, we will be concerned with the part of the motion before  $m_1$  reaches the pulley.



For questions 19 and 20, all assumptions remain except that  $M$  is no longer fixed to the floor but can move without friction on the floor.

19. By considering the motion (or non-motion) of the center of mass and/or otherwise, when  $m_2$  reaches the floor, what is the distance moved by  $M$ ? State in which direction if it has moved.

- a) 0
- b)  $\frac{m_1 h}{M + m_1 + m_2}$ , leftward
- c)  $\frac{m_1 h}{M + m_1 + m_2}$ , rightward
- d)  $\frac{m_1 h}{M + m_2}$ , leftward
- e)  $\frac{m_1 h}{M + m_2}$ , rightward

- 1 Horizontal external force is zero and the system starts at rest, so the horizontal CM position is fixed. As  $m_1$  moves right,  $M$  (with  $m_2$  pressed against it) moves left by some  $x$ .
- 2  $m_1$  moves  $h$  right *relative to*  $M$ , so relative to the ground it moves  $h - x$ .
- 3  $m_1(h - x) = (M + m_2)x \Rightarrow x = \frac{m_1 h}{M + m_1 + m_2}$ , leftward. **Answer (b)**

**Answer: b**

# 6 Circular Motion

Worked solutions to the examples in Chapter 6.

## 6.1 UNIFORM CIRCULAR MOTION IN A HORIZONTAL PLANE

### EXAMPLE 6.1 SJPO 2018 Q7

7. A bug is just about to slip on a circular turntable of radius  $R$  rotating at constant angular velocity. The bug is halfway between the centre and the edge and the coefficient of static friction is  $1/4$ . What is the acceleration of the bug?
- A.  $3g$   
B.  $2g$   
C.  $3g/2$   
D.  $g/2$   
E.  $g/4$

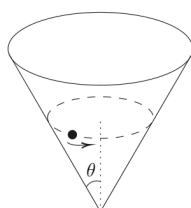
1 About to slip: static friction is at its maximum and supplies the centripetal force:  $ma_c = \mu_s N = \mu_s mg$ .

2  $a_c = \mu_s g = \frac{g}{4}$ . The radius doesn't matter. **Answer E**

**Answer: E**

### EXAMPLE 6.2 SJPO 2024 Q1

1. A smooth hollow cone of half-angle  $\theta$  is inverted with its axis vertical. A particle of mass  $m$  rolls around in a horizontal circle within the inner surface of the cone. What is the magnitude of the normal force  $N$  exerted by the cone on the particle?



- A.  $mg \cos \theta$   
B.  $mg \sin \theta$   
C.  $\frac{mg}{\cos \theta}$   
D.  $\frac{mg}{\sin \theta}$   
E.  $mg \tan \theta$

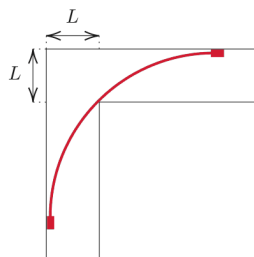
1 The wall makes angle  $\theta$  with the vertical, so  $N$  (perpendicular to the wall) makes angle  $\theta$  with the **horizontal**.

2 No vertical acceleration:  $N \sin \theta = mg \Rightarrow N = \frac{mg}{\sin \theta}$ . (The horizontal part  $N \cos \theta$  supplies  $mv^2/r$ .) **Answer D**

**Answer: D**

### EXAMPLE 6.3 SJPO 2024 Q18

18. A car drives at a constant speed along a horizontal road of width  $L$  that has a sharp  $90^\circ$  bend as shown in the figure. The coefficient of static friction between the car and the road is  $\mu$ . What is the maximum speed at which the car can negotiate the bend without slipping? Neglect the size of the car.



- A.  $\sqrt{\mu g L}$
- B.  $\sqrt{2\mu g L}$
- C.  $\sqrt{\sqrt{2}\mu g L}$
- D.  $\sqrt{(2 + \sqrt{2})\mu g L}$
- E.  $\sqrt{(2 + 2\sqrt{2})\mu g L}$

- 1 Use the largest arc that fits: tangent to both **outer** edges and just touching the **inner** corner. Put the outer corner at the origin, so the arc's centre is at  $(R, R)$  and the inner corner at  $(L, L)$ .
- 2 Inner corner on the arc:  $\sqrt{2}(R - L) = R \Rightarrow R = \frac{\sqrt{2}}{\sqrt{2} - 1}L = (2 + \sqrt{2})L$ .
- 3  $\frac{mv^2}{R} = \mu mg \Rightarrow v = \sqrt{(2 + \sqrt{2})\mu g L}$ . **Answer D**

**Answer: D**

## 6.2 VERTICAL AND NON-UNIFORM CIRCULAR MOTION

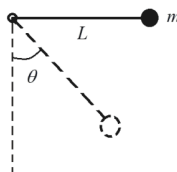
### EXAMPLE 6.4 SJPO 2022 Q10–11

Questions 10 and 11.

10. A simple pendulum, consisting of a string tied to a frictionless fixed pivot at one end and a bob of mass  $m$  at the other end, is gently released from the horizontal position (i.e.  $\theta = 90^\circ$ ) as shown.

After the release, the tension of the string is given by \_\_\_\_.

- a)  $3mg \cos \theta$
- b)  $3mg \sin \theta$
- c)  $mg \cos \theta$
- d)  $mg \sin \theta$
- e)  $4mg \cos \theta$



11. In order for the string not to break during the swing, it must be able to withstand a tension of \_\_\_\_.

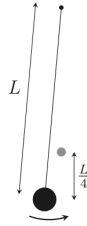
- a)  $2mg$
- b)  $4mg$
- c)  $8mg$
- d)  $6mg$
- e)  $3mg$

- 1 Energy: the bob has fallen  $L \cos \theta$ , so  $\frac{1}{2}mv^2 = mgL \cos \theta \Rightarrow \frac{mv^2}{L} = 2mg \cos \theta$ .
- 2 Radial:  $T - mg \cos \theta = 2mg \cos \theta \Rightarrow T = 3mg \cos \theta$ . **Q10 answer (a)**
- 3 Maximum at  $\theta = 0$  (lowest point):  $T_{\max} = 3mg$ . **Q11 answer (e)**

**EXAMPLE 6.5** SJPO 2026 Q14

**Question 14**

A pendulum of mass  $m$  and length  $L$  swings in the vertical plane. At the lowest point of its motion, the string catches on a peg located at a distance  $L/4$  above the mass. If the tension in the string just before hitting the peg is  $T = 3mg$ , what is the tension in the string right after hitting the peg? Neglect air resistance.



- (a)  $\frac{3}{4}mg$
- (b)  $\frac{3}{2}mg$
- (c)  $3mg$
- (d)  $\frac{5}{2}mg$
- (e)  $9mg$

- 1 Before:  $T - mg = \frac{mv^2}{L} = 2mg$ .
- 2 After: same  $v$ , radius  $L/4$ , so  $\frac{mv^2}{L/4} = 4(2mg) = 8mg$ .
- 3  $T' = mg + 8mg = 9mg$ . **Answer (e)**

**Answer: (e)**

# 7 Gravitation

Worked solutions to the examples in Chapter 7.

## 7.1 GRAVITATIONAL FORCE AND CIRCULAR ORBITS

### EXAMPLE 7.1 SJPO 2018 Q8

8. The height of a circular geosynchronous orbit (24 hr period) from the earth's surface is approximately \_\_\_\_ Earth radii. (Hint: You may need the gravitational constant, mass of the Earth and the gravitational acceleration at the surface of the Earth. These are given in the list of constants.)

- A. 3
- B. 4
- C. 5
- D. 6
- E. 7

1  $\frac{GMm}{r^2} = mr\omega^2$  with  $\omega = \frac{2\pi}{24 \times 3600} = 7.27 \times 10^{-5} \text{ rad s}^{-1}$ .

2  $r = \left( \frac{GM_E}{\omega^2} \right)^{1/3} = \left( \frac{(6.67 \times 10^{-11})(5.92 \times 10^{24})}{(7.27 \times 10^{-5})^2} \right)^{1/3} = 4.21 \times 10^7 \text{ m}$

3 Height =  $\frac{r - R_E}{R_E} = \frac{4.21 \times 10^7 - 6.37 \times 10^6}{6.37 \times 10^6} = 5.6 \approx 6$  Earth radii. **Answer D**

**Answer: D**

### EXAMPLE 7.2 SJPO 2026 Q24

#### Question 24

A satellite of mass  $m$  is in a circular orbit of radius  $r$  around a planet of mass  $M$ . It can be shown that its kinetic energy and potential energy are given by

$$\begin{aligned}\text{Kinetic energy} &= \frac{GMm}{2r} \\ \text{Potential energy} &= -\frac{GMm}{r}\end{aligned}$$

Suppose that the satellite's engine is suddenly shut down. As a result of resistive forces, how do the radius of the orbit and the linear speed of the satellite change over time?

- (a) Radius decreases and speed decreases.
- (b) Radius decreases and speed increases.
- (c) Radius increases and speed decreases.
- (d) Radius increases and speed increases.
- (e) Radius remains unchanged and speed decreases.

1  $E = K + U = \frac{GMm}{2r} - \frac{GMm}{r} = -\frac{GMm}{2r} = -K.$

2 Resistive forces decrease  $E$  (more negative), so  $r$  decreases and  $K = -E$  increases.

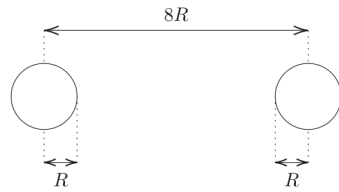
3 Radius decreases and speed increases. **Answer (b)**

**Answer: (b)**

## 7.2 GRAVITATIONAL POTENTIAL ENERGY

### EXAMPLE 7.3 SJPO 2024 Q23

23. In an experiment gone *very* wrong, Shuhao accidentally created a clone of Earth with the same mass and radius  $R$  as our Earth. The two Earths are initially stationary with a separation of  $8R$  between their centres. As they attract each other due to gravity, they eventually crash into each other. What is the velocity of each Earth at impact?



- A.  $\sqrt{\frac{1}{6}gR}$
- B.  $\sqrt{\frac{3}{8}gR}$
- C.  $\sqrt{\frac{3}{4}gR}$
- D.  $\sqrt{6gR}$
- E.  $\sqrt{12gR}$

- 1 Momentum is zero throughout, so both have equal speed  $v$ . They touch when the centres are  $2R$  apart.
- 2 Energy:  $-\frac{GM^2}{8R} = -\frac{GM^2}{2R} + 2 \cdot \frac{1}{2}Mv^2 \Rightarrow Mv^2 = \frac{3GM^2}{8R}$
- 3 With  $GM = gR^2$ :  $v = \sqrt{\frac{3GM}{8R}} = \sqrt{\frac{3}{8}gR}$ . **Answer B**

**Answer: B**

### EXAMPLE 7.4 SJPO 2026 Q23

#### Question 23

A mass is initially held at a height  $h$  above the Earth's surface, with the Sun directly overhead. The mass is then released. What is the minimum height  $h$  such that the mass proceeds to rise towards the Sun instead of falling back towards the Earth?

Data:

- Mass of Sun  $M_S = 1.99 \times 10^{30}$  kg
- Mass of Earth  $M_E = 5.97 \times 10^{24}$  kg
- Radius of Sun  $R_S = 6.96 \times 10^5$  km
- Radius of Earth  $R_E = 6.37 \times 10^3$  km
- Distance between the centres of the Sun and Earth  $d_{SE} = 1.50 \times 10^8$  km

- (a)  $4.43 \times 10^2$  km
- (b)  $2.59 \times 10^4$  km
- (c)  $2.53 \times 10^5$  km
- (d)  $1.49 \times 10^6$  km
- (e)  $7.50 \times 10^7$  km

- 1 The mass rises towards the Sun once the Sun's pull exceeds Earth's. Threshold at distance  $r$  from Earth's centre:  $\frac{GM_E}{r^2} = \frac{GM_S}{(d_{SE} - r)^2}$ .
- 2  $r = \frac{d_{SE}}{1 + \sqrt{M_S/M_E}} = \frac{1.50 \times 10^8}{1 + \sqrt{3.33 \times 10^5}} = \frac{1.50 \times 10^8}{578} = 2.59 \times 10^5$  km
- 3  $h = r - R_E = 2.59 \times 10^5 - 6.37 \times 10^3 = 2.53 \times 10^5$  km. **Answer (c)**

**Answer: (c)**

# 8 Rotational Dynamics

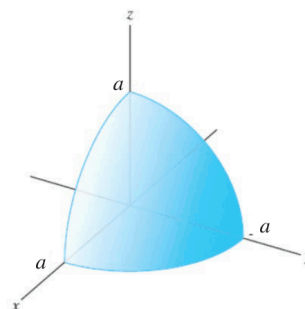
Worked solutions to the examples in Chapter 8.

## 8.1 MOMENT OF INERTIA AND TORQUE

### EXAMPLE 8.1 SJPO 2018 Q9

9. The moment of inertia of a solid sphere of mass  $M$  and radius  $a$ , about an axis passing through its centre, is  $(2/5)Ma^2$ . The mass of a solid uniform octant (one-eighth) of a sphere of radius  $a$ , is  $m$ . The moment of inertia about an axis along one of the straight edges (e.g.  $z$  axis) is \_\_\_\_.

- A.  $\frac{4}{3}ma^2$   
 B.  $\frac{2}{5}ma^2$   
 C.  $\frac{3}{8}ma^2$   
 D.  $\frac{1}{8}ma^2$   
 E.  $\frac{1}{20}ma^2$



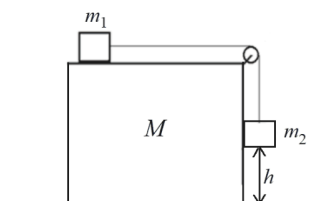
- 1 The straight edge lies along an axis through the sphere's centre. The 8 octants are identical and each has the same distribution of distances from this axis.
- 2 So  $I_{\text{octant}} = \frac{1}{8}I_{\text{sphere}} = \frac{1}{8} \cdot \frac{2}{5}(8m)a^2 = \frac{2}{5}ma^2$ . **Answer B**

**Answer: B**

### EXAMPLE 8.2 SJPO 2022 Q21

For questions 17- 21

A block of mass  $m_1$  is placed on top of a block of mass  $M$ . Holding block  $m_1$  stationary, it is joined by a string which passes over the pulley to a block of mass  $m_2$  hanging by the side of (and in contact with) block  $M$ , and is at height  $h$  above the floor. We will assume that the string is inextensible and massless, and all the contacts between objects in relative motion are frictionless. Furthermore, we will be concerned with the part of the motion before  $m_1$  reaches the pulley.



21. For this question, assume that  $M$  is fixed to the floor, and the mass and radius of the pulley is  $m_3$  and  $R$ , respectively. What is the acceleration of blocks  $m_1$  and  $m_2$ ? Hint: Take the moment of inertia of the pulley as  $I = \frac{1}{2}m_3R^2$

- a)  $\frac{m_1}{m_1+m_2+\frac{m_3}{2}}g$   
 b)  $\frac{m_2}{m_1+m_2+\frac{m_3}{2}}g$   
 c)  $\left(1 - \frac{m_3}{2(m_1+m_2)}\right)g$   
 d)  $\frac{m_1+m_2+\frac{m_3}{2}}{m_1}g$   
 e)  $\frac{m_1+m_2+\frac{m_3}{2}}{m_2}g$

- 1  $m_2: m_2g - T_2 = m_2a$ .  $m_1: T_1 = m_1a$ .
- 2 Pulley:  $(T_2 - T_1)R = \frac{1}{2}m_3R^2 \cdot \frac{a}{R} \Rightarrow T_2 - T_1 = \frac{1}{2}m_3a$ .
- 3 Add all three:  $m_2g = (m_1 + m_2 + \frac{m_3}{2})a \Rightarrow a = \frac{m_2}{m_1 + m_2 + \frac{m_3}{2}}g$ . **Answer (b)**

Answer: b

## 8.2 ROLLING WITHOUT SLIPPING

### EXAMPLE 8.3 SJPO 2018 Q4

4. A sphere rolls without slipping down a rough inclined plane. The gain in rotational kinetic energy is due **directly** to the work done by \_\_\_\_\_.
- A. Static friction
  - B. Kinetic friction
  - C. Weight
  - D. Normal contact force
  - E. Air resistance
- 1 About the centre of mass, weight and the normal force pass through the CM, so they exert no torque. Only **static friction** at the contact point exerts a torque about the CM.
- 2 So rotational KE is gained through friction's torque. Overall, friction does zero net work: it takes translational KE and converts it to rotational. **Answer A**

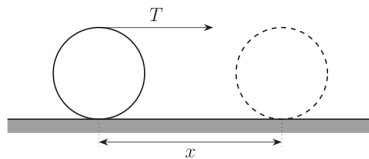
Answer: A

### EXAMPLE 8.4 SJPO 2026 Q17–18

Use the information below to answer Questions 17 and 18.

A uniform solid cylinder lies on rough horizontal ground. A string wrapped around the cylinder is pulled at the top with constant horizontal tension  $T$ . The cylinder rolls without slipping, moving a distance  $x$  forward.

(The moment of inertia of a uniform solid cylinder of mass  $m$  and radius  $r$  is  $I = \frac{1}{2}mr^2$  about its central axis.)



#### Question 17

What is the magnitude of the friction exerted on the cylinder by the ground?

- (a) 0
- (b)  $\frac{1}{4}T$
- (c)  $\frac{1}{3}T$
- (d)  $\frac{1}{2}T$
- (e)  $T$

#### Question 18

What is the work done by the tension  $T$ ?

- (a) 0
- (b)  $\frac{1}{2}Tx$
- (c)  $Tx$
- (d)  $\frac{3}{2}Tx$
- (e)  $2Tx$

- 1 Take friction  $f$  forward. Linear:  $ma = T + f$ . Rotation about the centre:  $\frac{1}{2}mr^2\alpha = Tr - fr$ .
- 2 Rolling:  $a = r\alpha$ , so  $\frac{1}{2}ma = T - f$ . Combining with  $ma = T + f$ :  $\frac{1}{2}(T + f) = T - f \Rightarrow f = \frac{1}{3}T$  (forward). **Q17 answer (c)**

- 3 Q18: the top of the cylinder moves at  $2v$ , so the string's end moves  $2x$ :  $W_T = T(2x) = 2Tx$ .  
(Equivalently  $Tx + Tr \cdot \frac{x}{r}$ .) **Q18 answer (e)**

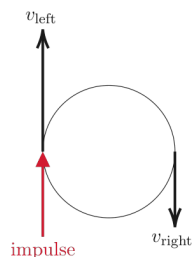
**Answer: Q17 (c), Q18 (e)**

### 8.3 ANGULAR MOMENTUM AND ANGULAR IMPULSE

#### EXAMPLE 8.5 **SJPO 2024 Q20**

20. A uniform solid disc rests flat on a frictionless table. A brief tangential impulse is applied on the leftmost point of the disc. What is the resulting ratio of speeds of the leftmost and rightmost points of the disc,  $\frac{|v_{\text{left}}|}{|v_{\text{right}}|}$ ?

(The moment of inertia of a uniform solid disc of mass  $M$  and radius  $R$  about its central axis is given by  $I = \frac{1}{2}MR^2$ .)



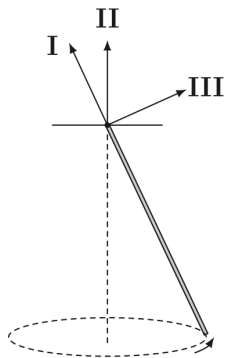
- A. 1
- B.  $\frac{5}{3}$
- C. 2
- D. 3
- E. 4

- 1 Linear:  $J = Mv$ . Angular about the CM:  $JR = \frac{1}{2}MR^2\omega$ .
- 2 Dividing:  $R\omega = 2v$ .
- 3 The leftmost point moves with the impulse:  $v + R\omega = 3v$ . The rightmost point:  $v - R\omega = -v$ .  
Ratio = 3. **Answer D**

**Answer: D**

Question 19

A thin stick is pivoted at its top end and is set to swing rigidly in uniform circular motion about the vertical axis. What are the directions of its angular velocity  $\vec{\omega}$  and its angular momentum  $\vec{L}$ ?



- (a)  $\vec{\omega}$  in direction **I** and  $\vec{L}$  in direction **I**
- (b)  $\vec{\omega}$  in direction **II** and  $\vec{L}$  in direction **I**
- (c)  $\vec{\omega}$  in direction **II** and  $\vec{L}$  in direction **II**
- (d)  $\vec{\omega}$  in direction **II** and  $\vec{L}$  in direction **III**
- (e)  $\vec{\omega}$  in direction **III** and  $\vec{L}$  in direction **II**

- 1  $\vec{\omega}$  lies along the rotation axis, which is vertical: direction **II**.
- 2 At the instant shown, each point has  $\vec{r}$  along the stick (downwards from the pivot) and  $\vec{p}$  into the page. So  $\vec{r} \times \vec{p}$  is perpendicular to the stick, in the plane of the page: direction **III**.
- 3  $\vec{\omega}$  along **II**,  $\vec{L}$  along **III**. **Answer (d)**

**Answer: (d)**

# 9 Fluids

Worked solutions to the examples in Chapter 9.

## 9.1 PRESSURE AND BUOYANCY

### EXAMPLE 9.1 SJPO 2022 Q24

24. How much weight can a Styrofoam cup of volume  $200 \text{ cm}^3$  carry before it sinks in water? Assume the weight of Styrofoam is negligible and take the density of water as  $1 \text{ g}\cdot\text{cm}^{-3}$ .

- a) 0.2N
- b) 5N
- c) 20N
- d) 1N
- e) 2N

- 1 Maximum upthrust when fully submerged:  $U = \rho V g = (1000)(200 \times 10^{-6})(9.8) = 1.96 \text{ N}$ .
- 2 The cup is massless, so it can carry about 2 N. **Answer (e)**

**Answer: e**

### EXAMPLE 9.2 SJPO 2022 Q25–26

For questions 25 and 26

The specific gravity of a substance is defined as the ratio of the density of the substance to that of pure water. The specific gravity of pure ice is 0.92, and that of seawater is 1.03.

25. What percentage of an iceberg made up of pure ice is above the surface of the seawater?

- a) 9%
- b) 8%
- c) 12%
- d) 11%
- e) 10%

26. A piece of pure ice is gently placed into a measuring cylinder containing seawater. The ice floats and the seawater level is noted. After the ice has melted, the seawater level will \_\_\_\_.

- a) rise
- b) stay the same
- c) drop
- d) depend on the amount of the pure ice, i.e. can be either a) or b) or c)

- 1 Floating:  $\frac{V_{\text{sub}}}{V} = \frac{0.92}{1.03} = 0.893$ , so  $10.7\% \approx 11\%$  is above the surface. **Q25 answer (d)**
- 2 Q26: the ice displaces seawater of volume  $\frac{m}{1.03\rho_w}$ . When melted it becomes pure water of volume  $\frac{m}{1.00\rho_w}$ , which is larger.
- 3 The meltwater needs more room than the hole it left, so the level **rises**. **Q26 answer (a)**

**Answer: Q25 d, Q26 a**

## 9.2 FLUID FLOW: CONTINUITY AND BERNOULLI

### EXAMPLE 9.3 SJPO 2022 Q22

22. Water is flowing vertically downward from a tap at a moderate speed. As it falls, the water column's diameter \_\_\_\_.

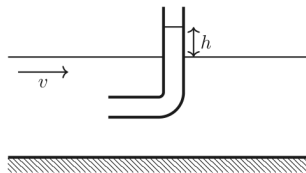
- a) decreases
- b) remains unchanged
- c) increases
- d) decreases in Northern Hemisphere but increases in Southern Hemisphere
- e) increases in Northern Hemisphere but decreases in Southern Hemisphere

- 1 Falling water speeds up under gravity, so  $v$  increases downward.
- 2 Continuity:  $Av = \text{constant}$ , so  $A$  decreases and the diameter **decreases**. The Coriolis options are irrelevant at this scale. **Answer (a)**

**Answer: a**

### EXAMPLE 9.4 SJPO 2024 Q24

24. An L-shaped pipe is placed into a river flowing at speed  $v$ , with the open end facing the current head-on. The water in the pipe is observed to rise to a height  $h = 0.50$  m above the water surface of the river. Find  $v$ .



- A.  $3.1 \text{ m s}^{-1}$
- B.  $4.2 \text{ m s}^{-1}$
- C.  $5.6 \text{ m s}^{-1}$
- D.  $6.2 \text{ m s}^{-1}$
- E.  $7.0 \text{ m s}^{-1}$

- 1 At the mouth the water is brought to rest (stagnation):  $p_{\text{mouth}} = p + \frac{1}{2}\rho v^2$ . This extra pressure supports a column of height  $h$ :  $\frac{1}{2}\rho v^2 = \rho gh$ .
- 2  $v = \sqrt{2gh} = \sqrt{2(9.8)(0.50)} = 3.1 \text{ m s}^{-1}$ . **Answer A**

**Answer: A**

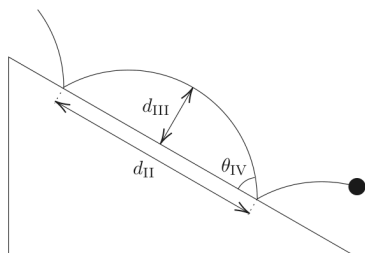
# Tutorial Solutions

Worked solutions to the 15 tutorial questions.

## QUESTION T1 SJPO 2024 Q10

10. A ball is dropped from rest onto an inclined slope. Assuming there are no energy losses, which of the following statements are true about the ball's motion as it bounces down the ramp?

- I. The time interval between each bounce decreases over time.
- II. The distance between each bounce,  $d_{II}$ , increases over time.
- III. The maximum distance reached by the ball from the ramp between each bounce,  $d_{III}$ , remains constant over time.
- IV. The angle of the ball's trajectory from the ramp at each bounce,  $\theta_{IV}$ , remains constant over time.



- A. I and II only
- B. II and III only
- C. III and IV only
- D. I, II and III only
- E. I, III and IV only

- 1 Use axes along ( $\parallel$ ) and perpendicular ( $\perp$ ) to the slope:  $g_{\perp} = g \cos \theta$ ,  $g_{\parallel} = g \sin \theta$  down the slope.
- 2  $\perp$  motion: each elastic bounce reverses  $v_{\perp}$  with the same magnitude, so every hop has the same flight time  $2v_{\perp}/g_{\perp}$  and the same maximum height from the ramp. I is false, III is true.
- 3  $\parallel$  motion:  $v_{\parallel}$  keeps increasing (it is never reversed), so each hop covers a longer distance along the slope. II is true. The angle to the ramp,  $\tan^{-1}(v_{\perp}/v_{\parallel})$ , therefore decreases. IV is false.
- 4 II and III only. **Answer B**

**Answer: B**

## QUESTION T2 SJPO 2026 Q8

### Question 8

A droplet falls through a tree and collides with a leaf every vertical distance  $h$ . Each collision causes the droplet to lose half of its speed. After falling through a large distance, the droplet achieves steady state. What is the terminal speed reached by the droplet? Neglect air resistance.

- (a)  $\sqrt{gh}$
- (b)  $\sqrt{2gh}$
- (c)  $\sqrt{\frac{2}{3}gh}$
- (d)  $\sqrt{\frac{8}{3}gh}$
- (e)  $\sqrt{8gh}$

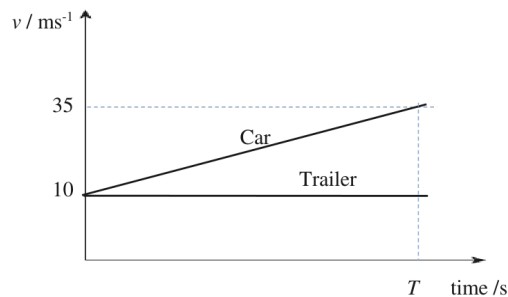
- 1 In the steady state, the droplet reaches the same speed  $v$  just before every collision and leaves each collision at  $v/2$ .
- 2 Free fall through  $h$  between leaves:  $v^2 = \left(\frac{v}{2}\right)^2 + 2gh \Rightarrow \frac{3}{4}v^2 = 2gh$ .
- 3  $v = \sqrt{\frac{8}{3}gh}$ . **Answer (d)**

**Answer: (d)**

### QUESTION T3 SJPO 2018 Q11

11. The minimum time  $T$  for a car to safely overtake a long trailer is measured from the time the front of the car is level with the rear of the trailer, until the rear of the car is one full car-length ahead of the trailer. The car is 3.5 m long and the trailer is 15.0 m long. The graph shows the speed-time graphs of the car and the trailer. What is the minimum time  $T$ ?

- A. 2.16 s
- B. 1.76 s
- C. 1.48 s
- D. 0.88 s
- E. 0.64 s



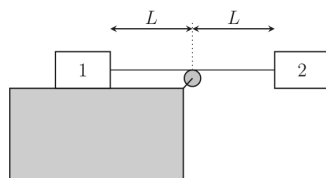
- 1 Relative to the trailer, the car's front must travel: trailer length + car length (to clear) + one more car length =  $15.0 + 3.5 + 3.5 = 22.0$  m.
- 2 Relative velocity starts at 0 (both at  $10 \text{ m s}^{-1}$ ) and rises linearly to  $35 - 10 = 25 \text{ m s}^{-1}$  at time  $T$ . Relative displacement is the area between the graphs:  $\frac{1}{2}(25)T$ .
- 3  $\frac{1}{2}(25)T = 22.0 \Rightarrow T = 1.76 \text{ s}$ . **Answer B**

**Answer: B**

### QUESTION T4 SJPO 2026 Q12

#### Question 12

Two blocks of equal masses are connected by an inextensible string passing over a light pulley at the edge of a table. Block 1 is on the table, while block 2 is suspended freely. Both blocks are initially held at equal horizontal distances from the pulley. The system is then released from rest. Which statement is true about the motion of the blocks? Neglect friction and air resistance.

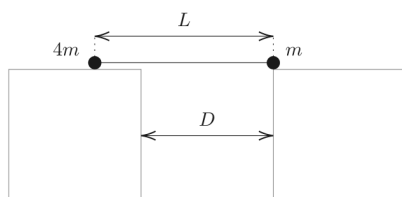


- (a) Both blocks remain at rest as their masses are equal.
  - (b) Block 1 reaches the pulley before block 2 strikes the table.
  - (c) Block 1 reaches the pulley after block 2 strikes the table.
  - (d) Block 1 reaches the pulley and block 2 strikes the table at the same time.
  - (e) Not enough information to determine which event occurs first.
- 1 Block 2 hangs at a horizontal distance from the pulley, so the string to it makes an angle  $\theta$  with the horizontal. The tension is the same throughout.
  - 2 Horizontal equations: block 1,  $ma_1 = T$ ; block 2,  $ma_2 = T \cos \theta \leq T$ .
  - 3 So  $a_1 \geq a_2$  at every instant. Both start from rest and must cover the same horizontal distance  $L$ , so block 1 gets there first. **Answer (b)**

Answer: (b)

QUESTION T5 SJPO 2024 Q16

16. Brian wants to bridge two cliffs using a plank of length  $L$ . The plank itself is essentially massless, but there is a mass  $m$  attached to its front and a mass  $4m$  attached to its back. The bridge is formed by pushing the plank horizontally across the gap, until the full gap is covered by the plank. What is the maximum gap  $D$  between the cliffs that can be bridged in this manner?



- A.  $\frac{1}{5}L$
- B.  $\frac{1}{2}L$
- C.  $\frac{3}{5}L$
- D.  $\frac{3}{4}L$
- E.  $\frac{4}{5}L$

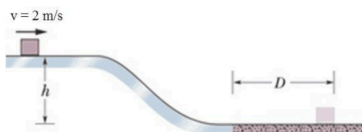
- 1 The plank (massless) carries  $4m$  at the back and  $m$  at the front. Its CM is  $\frac{mL}{5m} = \frac{L}{5}$  from the back mass.
- 2 While the plank is pushed out over the gap, it tips once its CM passes the near cliff edge. At the maximum gap, the front just reaches the far cliff when the CM is exactly at the edge.
- 3 Distance from the edge to the front  $= L - \frac{L}{5}$ , so  $D_{\max} = \frac{4}{5}L$ . **Answer E**

Answer: E

## QUESTION T6 SJPO 2022 Q14–16

For questions 14 – 16

A block of mass  $m$  slides with an initial velocity of 2 m/s along a horizontal track that descends through distance  $h$ . The track is frictionless except for the lower section. There the block slides to a stop in a certain distance  $D$  because of friction.



14. If  $h$  is halved,  $D$  \_\_\_\_.

- a) will be halved
- b) will be unchanged
- c) will be doubled
- d) will be tripled
- e) cannot be determined due to insufficient information

15. If  $m$  is doubled,  $D$  \_\_\_\_.

- a) will be halved
- b) will be unchanged
- c) will be doubled
- d) will be tripled
- e) cannot be determined due to insufficient information

16. If  $m$  is doubled, the total thermal energy generated due to friction as the block slides to a stop \_\_\_\_.

- a) is halved
- b) is unchanged
- c) is doubled
- d) is tripled
- e) cannot be determined due to insufficient information

1 Energy:  $\frac{1}{2}mu^2 + mgh = \mu_k mg D \Rightarrow D = \frac{u^2/2 + gh}{\mu_k g}$  (frictionless except the rough section).

2 Q14: halving  $h$  halves only the  $gh$  term. The new distance is  $D' = \frac{u^2/2 + gh/2}{\mu_k g}$ , and  $D'/D$  depends on  $h$ , which is not given. Strictly, the change **cannot be determined (e)**. The official key gives (a) "halved", which holds only if the initial KE is negligible ( $u^2 \ll 2gh$ ).

3 Q15:  $m$  cancels, so  $D$  is **unchanged. (b)**

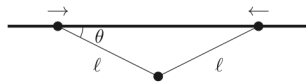
4 Q16: heat =  $\frac{1}{2}mu^2 + mgh \propto m$ , so doubling  $m$  **doubles it. (c)**

**Answer: Q14 e (key: a), Q15 b, Q16 c**

**About the answer key.** Q14: the official key says (a). That assumes the 2 m s<sup>-1</sup> initial speed contributes negligible energy. Taken literally,  $D$  is not exactly halved, and since  $h$  is not given the factor cannot be found, which is (e). Either could have been the setters' intent. What matters is that you can explain the difference.

## QUESTION T7 SJPO 2024 Q14

14. Three identical point masses  $m$  are connected by strings of length  $\ell$  in the vertical plane, as shown below. The two top masses are constrained to move horizontally along a frictionless track. If the system is released from rest at an initial angle  $\theta = 0$  from the horizontal, find the relative velocity of the two top masses just before they collide with each other.



- A. 0
- B.  $\sqrt{g\ell}$
- C.  $\sqrt{2g\ell}$
- D.  $\sqrt{3g\ell}$
- E.  $\sqrt{4g\ell}$

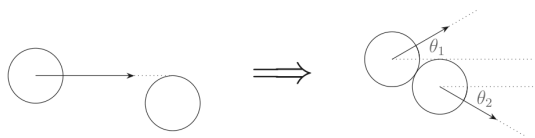
- 1 Just before the top masses collide, both strings are vertical. The bottom mass then has no horizontal velocity, by symmetry. Its velocity relative to each top mass must be perpendicular to the (vertical) string, so its vertical velocity is also zero. It is momentarily at rest.
- 2 The bottom mass has fallen  $\ell$ :  $mg\ell = 2 \cdot \frac{1}{2}mv^2 \Rightarrow v = \sqrt{g\ell}$  for each top mass.
- 3 Relative velocity  $= 2v = 2\sqrt{g\ell} = \sqrt{4g\ell}$ . **Answer E**

**Answer: E**

## QUESTION T8 SJPO 2026 Q16

### Question 16

Two identical billiard balls undergo a side-on elastic collision on a frictionless table, where one ball strikes another stationary ball off-centre. The moving ball is deflected by an angle  $\theta_1 = 75^\circ$ . What is the deflection angle  $\theta_2$  of the other ball?



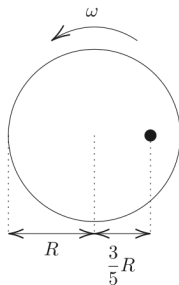
- (a)  $15^\circ$
- (b)  $30^\circ$
- (c)  $45^\circ$
- (d)  $60^\circ$
- (e)  $75^\circ$

- 1 Equal masses, elastic, one at rest:  $\vec{u} = \vec{v}_1 + \vec{v}_2$  and  $u^2 = v_1^2 + v_2^2$ .
- 2 Squaring the first and subtracting the second:  $2\vec{v}_1 \cdot \vec{v}_2 = 0$ , so the final velocities are perpendicular:  $\theta_1 + \theta_2 = 90^\circ$ .
- 3  $\theta_2 = 90^\circ - 75^\circ = 15^\circ$ . **Answer (a)**

**Answer: (a)**

## QUESTION T9 SJPO 2024 Q17

17. A horizontal turntable of radius  $R$  rotates with constant angular velocity  $\omega$ . An ant, positioned a distance  $\frac{3R}{5}$  from the centre, initially rotates along with the turntable. All of a sudden, it decides to jump vertically (relative to itself). What is the minimum duration of its jump such that it escapes the turntable?



- A.  $\frac{2}{5\omega}$
- B.  $\frac{3}{5\omega}$
- C.  $\frac{4}{5\omega}$
- D.  $\frac{4}{3\omega}$
- E.  $\frac{5}{3\omega}$

- 1 When the ant leaves the surface, there is no horizontal force on it. It moves in a straight line along the **tangent** with speed  $v = \omega \cdot \frac{3}{5}R$ .
- 2 Distance along the tangent from its position to the rim:  $\sqrt{R^2 - \left(\frac{3}{5}R\right)^2} = \frac{4}{5}R$ .
- 3  $t = \frac{\frac{4}{5}R}{\frac{3}{5}\omega R} = \frac{4}{3\omega}$ . **Answer D**

**Answer: D**

## QUESTION T10 SJPO 2026 Q20

### Question 20

A mass is placed on the edge of a horizontal turntable of radius  $R$  and coefficient of static friction  $\mu_s$ . Starting from rest (at time  $t = 0$ ), the turntable begins to rotate at a constant angular acceleration  $\alpha$ . At what time  $t$  does the mass slip off the turntable?

- (a)  $\frac{\mu_s g}{R\alpha}$
- (b)  $\left(\frac{\mu_s g}{R\alpha^2}\right)^{1/2}$
- (c)  $\left(\frac{\mu_s g}{R\alpha^2} - \frac{1}{\alpha}\right)^{1/2}$
- (d)  $\left(\frac{\mu_s^2 g^2}{R^2 \alpha^2} - 1\right)^{1/2}$
- (e)  $\left(\frac{\mu_s^2 g^2}{R^2 \alpha^4} - \frac{1}{\alpha^2}\right)^{1/4}$

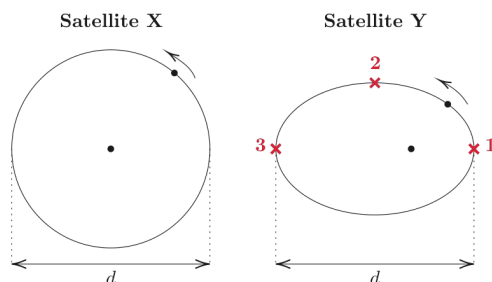
- 1 Static friction supplies both the tangential force  $mR\alpha$  and the centripetal force  $mR\omega^2 = mR\alpha^2 t^2$ , which are perpendicular.
- 2  $f = mR\sqrt{\alpha^2 + \alpha^4 t^4}$ . It slips when  $f = \mu_s mg$ .
- 3  $\alpha^2 + \alpha^4 t^4 = \frac{\mu_s^2 g^2}{R^2} \Rightarrow t = \left(\frac{\mu_s^2 g^2}{R^2 \alpha^4} - \frac{1}{\alpha^2}\right)^{1/4}$ . **Answer (e)**

**Answer: (e)**

## QUESTION T11 SJPO 2024 Q22

22. Two satellites X and Y orbit the same planet and have equal masses. The orbit of satellite X is circular, whereas the orbit of satellite Y is elliptical. The major axis of satellite Y's orbit has the same length as the diameter  $d$  of satellite X's orbit. Which of the following statements are true?

- I. The period of the orbit is equal for both satellites.
- II. The total energy of the orbit is equal for both satellites.
- III. The angular momentum of the orbit (with respect to the planet) is equal for both satellites.
- IV. The speeds of Satellite Y at points 1, 2 and 3 of its orbit (as shown below) are related by  $v_1 < v_2 < v_3$ .



- A. III only
- B. I and II only
- C. I and IV only
- D. III and IV only
- E. II, III and IV only

- 1 I: Kepler's third law uses the semi-major axis:  $T^2 = \frac{4\pi^2 a^3}{GM}$ . Equal  $a = d/2$  gives equal periods. **True.**
- 2 II: orbital energy depends only on  $a$ :  $E = -\frac{GMm}{2a}$ . **True.**
- 3 III: for a given energy, the circular orbit has the *largest* angular momentum. The ellipse has less. **False.**
- 4 IV: by energy conservation, the satellite is fastest nearest the planet (point 1) and slowest farthest away (point 3):  $v_1 > v_2 > v_3$ . **False.** I and II only. **Answer B**

**Answer: B**

## QUESTION T12 SJPO 2026 Q22

### Question 22

A tiny dust particle is at rest in space at a large distance  $r$  from the Earth. Due to Earth's gravitational attraction, the dust particle begins moving in a straight line towards the Earth. The Earth has mass  $M$  and radius  $R$ , with  $r \gg R$ . What is the time taken for the dust particle to reach the Earth?

- (a)  $\sqrt{\frac{2r}{g}}$
- (b)  $\sqrt{\frac{2r^3}{GM}}$
- (c)  $\frac{\pi}{2\sqrt{2}} \sqrt{\frac{r^3}{GM}}$
- (d)  $\pi \sqrt{\frac{r^3}{GM}}$
- (e)  $2\pi \sqrt{\frac{r^3}{GM}}$

- 1 Treat the straight-line fall as the limit of a very thin ellipse with the Earth at one focus. Its semi-major axis is  $a = r/2$ .

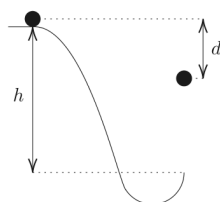
- 2 By Kepler's third law, its period equals that of a circular orbit of radius  $r/2$ :  $T = 2\pi\sqrt{\frac{(r/2)^3}{GM}} = \pi\sqrt{\frac{r^3}{2GM}}$ .
- 3 The fall from apoapsis to the Earth is half an orbit:  $t = \frac{T}{2} = \frac{\pi}{2\sqrt{2}}\sqrt{\frac{r^3}{GM}}$ . **Answer (c)**

**Answer: (c)**

### QUESTION T13 SJPO 2024 Q21

21. A uniform solid sphere is placed at the top of a valley and released from rest. It rolls down, and upon reaching the end of the valley, a height  $h$  below the top of the valley, it rises upward into the air. Throughout its motion, the sphere never experiences any energy dissipation and never slips. Yet, Chris claims that the sphere will never return to its original height, falling short by a distance  $d$ . Is Chris right, and if so, what is  $d$ ?

(The moment of inertia of a uniform solid sphere of mass  $M$  and radius  $R$  about its central axis is given by  $I = \frac{2}{5}MR^2$ .)



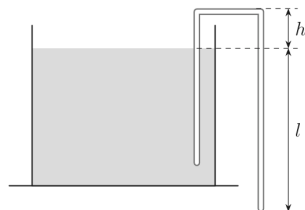
- A. Chris is wrong, as the sphere should return to its original height.
- B. Chris is right, but there is not enough information to calculate  $d$ .
- C. Chris is right,  $d = \frac{1}{10}h$ .
- D. Chris is right,  $d = \frac{2}{7}h$ .
- E. Chris is right,  $d = \frac{2}{5}h$ .
- 1 Rolling without slipping:  $K_{\text{rot}} = \frac{1}{2} \cdot \frac{2}{5}MR^2\omega^2 = \frac{2}{5}K_{\text{trans}}$ , so rotation holds  $\frac{2}{7}$  of the total  $Mgh$ .
- 2 Once airborne, no torque acts, so the sphere keeps spinning. Only its translational KE ( $\frac{5}{7}Mgh$ ) converts back into height.
- 3 It rises to  $\frac{5}{7}h$ , falling short by  $d = \frac{2}{7}h$ . Chris is right. **Answer D**

**Answer: D**

## QUESTION T14 SJPO 2026 Q21

### Question 21

A siphon is a seemingly gravity-defying device that can empty a tank of fluid by causing fluid to flow up, then down, and out of the siphon. In this setup, the siphon reaches a maximum height  $h$  above the fluid surface, and minimum height  $l$  below the fluid surface. Let  $P_0$  be the atmospheric pressure and  $\rho$  be the fluid density. What is the maximum height  $h$  for which the siphon will work? Assume that the siphon has a uniform diameter and that the tank is very large.



- (a)  $\frac{P_0}{\rho g}$
- (b)  $\frac{P_0}{2\rho g}$
- (c)  $\frac{P_0}{\rho g} - l$
- (d)  $\frac{P_0}{2\rho g} - l$
- (e) There is no maximum height; the siphon will always work for any  $h$ .

- 1 Bernoulli from the tank surface (large tank,  $v \approx 0$ ) to the outlet,  $l$  below:  $P_0 + \rho gl = P_0 + \frac{1}{2}\rho v^2 \Rightarrow \frac{1}{2}\rho v^2 = \rho gl$ .
- 2 Uniform tube, so the speed at the top is also  $v$ . Bernoulli from the surface to the top,  $h$  above:  $P_0 = p_{\text{top}} + \frac{1}{2}\rho v^2 + \rho gh$ .
- 3 Pressure cannot fall below zero:  $p_{\text{top}} = P_0 - \rho g(l + h) \geq 0 \Rightarrow h \leq \frac{P_0}{\rho g} - l$ . **Answer (c)**

**Answer: (c)**

## QUESTION T15 SJPO 2026 Q5

### Question 5

A bubble of uniform density  $\rho_b$  and volume  $V$  is deep within a fluid of uniform density  $\rho_f$ . What is the total change in potential energy of the system if the bubble rises by a small height  $h$ ?

- (a)  $\rho_b Vgh$
- (b)  $\rho_f Vgh$
- (c)  $(\rho_b - \rho_f) Vgh$
- (d)  $(\rho_f - \rho_b) Vgh$
- (e)  $(\rho_b + \rho_f) Vgh$

- 1 As the bubble rises by  $h$ , an equal volume  $V$  of fluid moves down by  $h$  to fill the space it left.
- 2  $\Delta U = \rho_b Vgh - \rho_f Vgh = (\rho_b - \rho_f) Vgh$ , which is negative when  $\rho_b < \rho_f$ , so rising is energetically favourable. **Answer (c)**

**Answer: (c)**